

Bushra Khan

M.Sc. Bioinformatics Candidate | RNA-seq | Structural Bioinformatics |

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Profile Summary

Enthusiastic and curious Bioinformatics postgraduate with early experience in RNA-seq analysis and an interest in protein structure prediction and computational approaches in biology. I've explored tools and workflows related to drug discovery and multi-omics data, and I'm actively building my skills in these areas. Eager to grow in the field by contributing to real-world research and learning through hands-on work in translational bioinformatics and personalized medicine.

Education

M.Sc. in Bioinformatics, Jamia Hamdard University, New Delhi | 2023 – Present

Relevant Coursework: Genomics, Transcriptomics, Structural Bioinformatics, Proteomics, Systems Biology, Machine Learning, Python and R Programming

B.Sc. in Biotechnology, Mohanlal Sukhadia University, Udaipur | 2019 – 2022

Research Interests

- Structural Bioinformatics
- RNA-seq and Differential Gene Expression
- AI/ML in Drug Discovery
- Protein Function Prediction
- Multi-omics Data Integration

Research Internship

Research Intern – School of Computational and Integrative Sciences, JNU | July 2025 – Current (Collaborative project with NIPGR)

- Performing HMMER, CDD, and multiple sequence alignment (MSA) searches on extracted genomic sequences for evolutionary insights.

- Conducting phylogenetic tree construction, synteny analysis, and mutation analysis using MCScanX.
- Applying bootstrapping approaches to identify variation among gene duplicates and infer evolutionary trajectories.

Research Intern – Bioclues.org | Jan 2025 – July 2025

- Benchmarked four AI-based tools (AlphaFold3, RosettaFold, ESMFold, Chai-1) on 39 human hypothetical proteins.
- Automated benchmarking workflows using Python; evaluated protein structures via RMSD, TM-score, GDT-TS, and pLDDT.
- Annotated structures using Foldseek, ProSA, VERIFY 3D, TMHMM, WoLF PSORT, Protein-Sol, and DAVID, etc..

Academic Projects

1. Functional and Structural Annotation of 1000 Hypothetical Proteins

- Developed a Python-based pipeline for structural prediction and function inference.
- Applied validation and annotation tools to evaluate protein models.

2. NGS RNA-seq Analysis

- Performed differential gene expression analysis using DESeq2 in R.
- Visualized results using volcano and MA plots.

Workshops and Technical Training

- NGS - Sequence to Insights (OmicsLogic): RNA-seq.
- AI & ML for Drug Discovery: QSAR modeling, GROMACS, ChimeraX
- Advance Internship (Rapture Biotech): Proteomics and Bioinformatics techniques
- Edufabrica–IIT Kharagpur Workshop: Genetics, endocrinology, molecular biology

Certifications

- Pre-Conference Workshop on Computational Drug Discovery: Bench to Market
- Mastering R for Bioinformatics – 3-Hour Workshop (DrOmics Labs)

Technical Skills

Programming Languages: Python, R, Bash, Perl

Bioinformatics Tools: DESeq2, Foldseek, BLAST, GROMACS, ChimeraX

Genomic Data Analysis: RNA-seq, CNV, Databases: UniProt, GenBank, ClinVAR
Visualization: ggplot2, Matplotlib, Seaborn
Platforms: Linux, GitHub

Publications & Presentations

- Bushra Khan & Tansuk Barupal, (2019). Water Hyacinth in slowing down eutrophication. National Journal of Life Sciences
- Presentation: Biofuels: Fuels of the Future, International Conference – MLSU, Udaipur
- Presentation: Structural Annotation of Hypothetical Proteins, Departmental Forum – Jamia Hamdard University

Leadership & Outreach

- Content Board Co-Head, Ignite Society – Led science communication & events
 - o Co-organised LinkedIn & Career Development Workshop with LedBy Foundation at U.S. Embassy, New Delhi
- Built public GitHub pipelines for RNA-seq and protein annotation projects

Soft Skills & Languages

Languages: English (Fluent), Hindi (Native), Japanese (Beginner)

Soft Skills: Scientific Writing, Teamwork, Leadership, Public Speaking, Time Management

References

Dr. Atul Kumar Jaiswal – Assistant Professor, Department of Computer Science and Engineering, School of Engineering Science and Technology, Jamia Hamdard (atuljaiswal@jamiahamdard.ac.in; atulj91@gmail.com)

Dr. Renuka Suravajhala – Assistant Professor, Amrita Vishwa Vidyapeetham; Co-founder, Bioclues.org (rainprash@gmail.com)

Mr. Bilal Ahmed Abbasi – Senior Data Analyst, TCS Research and Innovation Labs; Research Fellow & Project Mentor, Bioclues.org (bilahmd249@gmail.com)

Dr. Abhinaya Manivannan – Assistant Professor, School of Computational and Integrative Sciences, Jawaharlal Nehru University (abinayamanivannan@gmail.com)

